

DATABASE SYSTEMS

CHAP. 10: Introduction to Distributed Database Systems

Distributed Database Systems

- Definition

- A database that is under the control of a central DBMS in which storage devices are not all attached to a common CPU.

- Example

- A travel company with many branches across the country

Distributed Data Independence

- Users should not have to know where data is located.

Distributed Transaction Atomicity

- Users should be able to write transactions accessing multiple sites just like local transactions.

Recent Trends

- Users have to be aware of where data is located
- And distributed transaction atomicity are not supported.

Types of Distributed Databases

- Homogeneous
- Heterogeneous

Homogeneous

- Every site runs the same type of DBMS.
- All sites use the same DBMS product.
- They are Much easier to design and manage.
- This Approach provides incremental growth and allows increased performance.

Heterogeneous

- Different sites run different DBMSs (different RDBMSS or even nonrelational DBMSs).
- Sites may run different DBMS products, with possibly different underlying data models.

Heterogeneous(Cont.)

- Occurs when sites have implemented their own databases and integration is considered later.
- Translations required to allow for:
 - Different hardware.
 - Different DBMS products.
 - Different hardware and different DBMS products.
- Typical solution is to use gateways.

Distributed Database Architectures

- Client-Server
 - Communication duties are shared between the client machines and servers.
- Peer-to-peer systems
 - No distinction of client machines versus servers.
- Non-distributed systems

Functions of a DDBMS

- Functionality of a DBMS.
- Extended communication services.
- Extended data dictionary.
- Distributed query processing.
- Extended concurrency control.
- Extended recovery services.

Applications

- Manufacturing - especially multi-plant manufacturing
- Military command and control
- Corporate MIS
- Airlines
- Hotel chains
- Any organization which has a decentralized organization structure